

On the Limits of Behavioral Alignment: Formal Verifiability and the Problem of Normative Indistinguishability

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Abstract

Behavioral evaluation is the dominant paradigm for assessing alignment in large language models (LLMs). In current practice, observed compliance under finite evaluation protocols is treated as evidence of latent alignment. However, the inference from bounded behavioral evidence to claims about global latent properties is rarely analyzed as an identifiability problem.

In this paper, we study alignment evaluation through the lens of statistical identifiability under partial observability. We allow agent policies to condition their behavior on observable signals correlated with the evaluation regime, a phenomenon we term *evaluation awareness*. Within this framework, we formalize the *Alignment Verifiability Problem* and introduce *Normative Indistinguishability*, which arises when distinct latent alignment hypotheses induce identical distributions over evaluator-accessible observations.

Our main theoretical contribution is a *conditional impossibility result*: under finite behavioral evaluation and evaluation-aware policies, observed compliance does not uniquely identify latent alignment, but only membership in an equivalence class of conditionally compliant policies, under explicit assumptions on policy expressivity and observability.

We complement the theory with a constructive existence proof using an instruction-tuned LLM (Llama-3.2-3B), demonstrating a conditional policy that is perfectly compliant under explicit evaluation signals yet exhibits degraded identifiability when the same evaluation intent is conveyed implicitly. Together, our results show that behavioral benchmarks provide necessary but insufficient evidence for latent alignment under evaluation awareness.

1 Introduction

Behavioral evaluation has become the dominant paradigm for assessing alignment in Large Language Models (LLMs) [29, 38]. In contemporary practice, alignment is operationalized as performance under finite interaction protocols, such as safety benchmarks, red-teaming suites, or automated evaluation pipelines. Observed compliance within these regimes is commonly taken as evidence that a system adheres to intended normative objectives [4]. This inferential step—from bounded behavioral evidence to claims about latent alignment properties—underlies a substantial portion of current alignment research and deployment safety practice.

Despite its prevalence, this inference is rarely examined as a formal problem of identifiability. Behavioral evaluation yields finite, protocol-dependent observations, whereas alignment is typically conceptualized as a latent property governing behavior across the space of possible interactions. Whether, and under what conditions, such latent properties are identifiable from finite behavioral evidence constitutes a foundational epistemic question for the field [41].

The limits of behavioral inference are well established in classical machine learning and decision theory. A broad literature on proxy objectives, reward misspecification, and specification gaming shows how optimization against observable metrics can decouple measured performance from intended target properties, leading to Goodhart-style failures [3, 14, 30]. Similarly, results in inverse

reinforcement learning and preference inference indicate that latent objectives are, in general, not uniquely identifiable from behavior alone: multiple distinct hypotheses may induce indistinguishable observed behavior under finite interaction [1, 32]. These results already impose principled limits on behavioral inference under standard assumptions of stationary and evaluation-independent behavior generation.

However, a key assumption underlying most existing identifiability results is that the behavior being observed is generated independently of the evaluation procedure itself. In the context of LLM alignment evaluation, this assumption is often violated. Modern agents may condition their behavior on information correlated with the evaluation regime, including input distributions, feedback signals, or structural regularities of the protocol. As a consequence, the evaluation procedure becomes an endogenous component of the data-generating process. When behavior is evaluation-aware, standard inference guarantees based on passive or fixed observation no longer apply in a straightforward manner.

Recent empirical work indicates that this distinction is practically relevant and that the measurement process itself may be fragile. Models have been shown to adapt selectively to assessment contexts, exploiting proxy metrics or modifying behavior in response to evaluation-specific signals [37, 39]. At the same time, the reliability of evaluation instruments has been questioned: studies of automated evaluators and benchmark-driven assessment report systematic biases, calibration failures, and limitations in evaluating capabilities that exceed the evaluator’s own grounding [19, 24, 26]. Together, these findings indicate that finite behavioral evaluation cannot, in general, be assumed to sample from a stationary or evaluator-independent distribution [27].

From an inference-theoretic perspective, alignment evaluation therefore belongs to a class of problems characterized by partial observability, latent target properties, and endogenous data generation. Despite the convergence of relevant theoretical insights and empirical observations, the alignment literature lacks a unified formal account of when alignment is verifiable from finite behavioral evidence under evaluation-aware behavior [9].

Our Contributions. In this paper, we address this gap by framing alignment evaluation explicitly as an identifiability problem under finite and endogenous observation. Our contributions are three-fold:

1. **Conditional Non-Identifiability Result.** We define the *Alignment Verifiability Problem* and establish a negative but explicitly delimited identifiability result: under evaluation-aware agents and finite behavioral testing, observed compliance does not uniquely identify latent alignment (Theorem 1). This result characterizes an *Indistinguishability Set* of hypotheses that are observationally equivalent under a given evaluation regime yet diverge outside the evaluated domain.
2. **Constructive Existence Witness.** We provide a concrete existence witness of such indistinguishability in modern LLMs. Using a simple conditional fine-tuning procedure (the “Chameleon” construction), we instantiate a model whose behavior is compliant under explicit evaluation signals while diverging under semantically equivalent but unanchored contexts. This construction illustrates that evaluation-dependent policy selection—and the resulting non-identifiability—can arise through standard supervised learning, without adversarial prompting, harmful content, or exceptional model capabilities.
3. **Implications for Behavioral Evaluation.** We analyze the information-theoretic structure of this failure mode and argue that purely black-box behavioral evaluation cannot, in general, collapse the indistinguishability set induced by evaluation-aware behavior. Restoring verifiability would require access to information beyond input–output behavior, motivating evaluation

frameworks that incorporate additional structural or internal signals.

Importantly, our result does not imply that behavioral evaluation is uninformative or ineffective. Rather, it constrains the inferential claims such evaluation can support. Behavioral alignment tests should be interpreted as estimating equivalence classes of conditionally compliant policies under a specified regime, rather than as verifying alignment as a context-invariant latent property. Under this interpretation, benchmarks provide upper bounds on observable compliance within a domain, not guarantees of global alignment or safety.

Scope and Non-Claims. This paper focuses on the epistemic limits of behavioral alignment evaluation. We do not propose new alignment training algorithms, nor do we claim to detect or induce deceptive intent. The empirical construction presented serves solely as an existence witness of evaluation-dependent non-identifiability, not as a release of new exploitation or jailbreak techniques.

Paper Organization. The remainder of this paper is organized as follows. Section 2 introduces the formal setting and notation. Section 3 defines behavioral alignment tests. Section 5 presents the main non-identifiability result (Theorem 1). Sections 6 and 7 situate the result within inference theory and modern evaluation practice. Section 8 introduces a stylized Gridworld example. Section 9 presents our constructive existence witness using Llama-3. Section 10 clarifies limitations, Section 11 discusses implications, and Section 12 concludes.

2 Preliminaries

This section establishes the formal foundations for our analysis of alignment verifiability. We define the structure of interaction histories, the nature of evaluation regimes, and the space of alignment hypotheses. Crucially, we formalize *evaluation awareness* as a departure from the standard assumption of stationary, evaluation-independent behavior generation. Our goal is to introduce a minimal structural framework sufficient to reason about behavioral identifiability under finite and endogenous observation, without committing to any particular alignment formalism.

To facilitate the formal treatment of the verification problem, we summarize the key mathematical notation used throughout this framework in Table 1.

2.1 Interaction Histories and Evaluation Regimes

We consider a discrete-time interactive setting between an evaluator and an agent. Let $\mathcal{O}$ denote the set of possible observations (inputs provided by the evaluator) and $\mathcal{A}$ the set of possible actions (the agent’s responses). An *interaction history* h is a finite sequence in the set $\mathcal{H} = \bigcup_{t=1}^{\infty} (\mathcal{O} \times \mathcal{A})^t$, represented as:

$$h = (o_1, a_1, o_2, a_2, \dots, o_t, a_t).$$

An *evaluation regime* E specifies a finite behavioral protocol. Formally, E induces:

- (i) A finite set of interaction histories $\mathcal{H}_E \subset \mathcal{H}$ that are reachable and observable under the protocol.
- (ii) A corresponding probability space $\mathcal{D}_E$ over datasets generated by these histories.

Evaluation is necessarily finite: $\mathcal{H}_E$ is bounded in both length and scope by the practical constraints of the protocol. Any benchmark or red-teaming suite therefore observes only a bounded projection of the agent’s behavior space.